

eSamithi Member Mobile App — UI/UX & Motion Refresh

Status: Draft for approval · **Date:** 21 September 2026 · **App:** eSamithi Member 1.3.1 (Android versionCode 6) **Owner:** Sachira · **Applies to:** `mobile/` (Expo SDK 57, React Native 0.86) **Depends on:** `MEMBER-MOBILE-APP-REQUIREMENTS.md` (what the app does) · `ESAMITHI-WEB-APP-REQUIREMENTS.md` §4 (the shared design system this aligns to)

0. Executive summary

The member app was described by its owner as looking "unprofessional", with animations that "dance". This document specifies the fix.

The app is **not** badly built. It already has a token module, a genuinely sophisticated language-aware type system, and a disciplined shared component kit. What it has is three specific, measurable defects, and a design system that was only ever half-applied.

1 • The dancing is a library regression, not a design choice. Reanimated 4 changed the default spring from `{damping:10, mass:1, stiffness:100}` to `{damping:120, mass:4, stiffness:900}`. That new default is *critically damped* — it does not bounce. But the app passes **partial** spring configs, so `mass` silently stays at 4 while `damping` keeps a value chosen for the old library. Every spring in the app is therefore running at a damping ratio between 0.12 and 0.25, which means 44–69 % overshoot and 1.6–2.3 seconds of visible wobble. A tap on any button in the app currently springs 1.3 % *larger* than its resting size and oscillates for a second and a half. The active tab pill peaks at **166** % of its final size. This is the whole of the "dancing", and it is fixed by stating three numbers instead of two.

2 • The default language renders wrong. Sinhala is the default language. 27 places set `fontWeight` without a font family, which on Android silently falls back to the system Sinhala face at *regular* weight — so most headings in the app are not bold for most users, and the same places set no line height, so Sinhala's tall glyph stack crowds. Three of the six Sinhala tab labels do not fit their slot and ellipsize: the primary navigation is truncated in the language most members read.

3 • Six of the palette's colour pairs fail WCAG AA, including the primary blue both as a text colour (3.68:1) and as a button fill under white text (3.68:1).

On top of those: 19 distinct font sizes for a six-step scale, roughly 50 % token adoption, eight separate implementations of "a row inside a card" at six different heights, four parallel status-badge colour maps, and no `FlatList` anywhere in an app that renders a member's entire payment history into one scroll view.

What this document specifies: a corrected motion system with every value stated and checked; the mobile expression of the shared eSamithi design system; a raised legibility floor for an elderly, Sinhala-reading, low-end-Android audience; and a six-phase delivery where every phase ships on its own over the air.

What it deliberately does not do: change the navigation, the six tabs, the features, or the API. This is a refresh, not a rebuild.

1. Purpose, scope and decisions

1.1 Who this app is for

Members of a Sri Lankan village welfare society (a *maranadhara samithi* — a death-donation society). They are largely elderly and rural, on budget Android phones, often on 3G, often reading in sunlight. They open the app to answer four questions: *do I owe anything*, *what have I paid*, *what do I owe on my loan*, and *has anyone died*. A small classifieds marketplace (Puruka) sits alongside.

Design consequence: this is not an app that needs to be exciting. It needs to be legible, calm and trustworthy — the qualities of a passbook, not of a consumer app.

1.2 Decisions taken with the owner

#	Decision	Choice	Consequence
D1	How far to go	Refresh	Navigation, tabs, routes and features are untouched. Motion is fixed, the token migration is finished, and the worst screen compositions are rebuilt.
D2	Visual direction	Match eSamithi web/desktop	One brand across all three surfaces: the same blue scale, the same neutrals, the same type rules, the same motion vocabulary.
D3	Legibility	Raise the floor	Larger base type, ≥ 48 dp targets, fewer items per screen, and real support for the phone's own font-size setting. Less fits per screen; that is the intended trade.
D4	Release route	Over-the-air first	Everything in phases 1–5 ships on the <code>prod</code> EAS channel. The app icon, splash, adaptive icon and notification colour are deferred to the next planned binary (phase 6).

1.3 Out of scope

The tab structure and information architecture · any API or data-model change · the desktop and web apps · the Puruka feature set · push notification behaviour · offline support · the app icon and splash (deferred to phase 6, not cancelled).

1.4 Non-negotiables

1. **No regression in what the app can do.** Every screen keeps its function.
 2. **Sinhala is the default and is tested first.** Any change verified only in English is not verified.
 3. **Every phase is shippable.** A half-finished refresh must never reach a member.
 4. `npx tsc --noEmit` **stays green** — it is the only automated gate this project has.
 5. **Three Android traps stay fixed.** SVG gradients must remain siblings of their Pressable, not children; Android `elevation` stays off large-radius cards; no `fontWeight` without a matching family.
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2. Current state — the evidence

Every claim here was verified against the working tree on 21 September 2026 and cites `file:line`. Contrast ratios were computed, not estimated.

2.1 Motion

Reanimated 4.5.0's `withSpring` default is `GentleSpringConfig = {damping:120, mass:4, stiffness:900}` (`node_modules/react-native-reanimated/src/animation/spring/springConfigs.ts:28`), merged with `{duration:550, dampingRatio:1}`. On its own that is critically damped: $\zeta = 1.00$, settling in ~267 ms with no overshoot. The previous major version defaulted to `{damping:10, mass:1, stiffness:100}` (same file, line 6).

The app was written against the old defaults and passes partial configs. Because `safeMergeConfigs` shallow-merges, an unspecified `mass` stays 4:

Where	As written	ζ	Overshoot	Settle
<code>src/ui/pressable.tsx:38,42</code> — every tap, 33 call sites	<code>{damping:20, stiffness:400}</code>	0.25	44 %	1.6 s
<code>src/app/(tabs)/index.tsx:20</code> — Home, 7 staggered sections	<code>.springify().damping(18)</code>	0.15	62 % (15.5 px past its mark)	1.8 s
<code>src/app/(auth)/index.tsx:36,44,50,56,81</code> — Welcome, 5 elements	<code>.springify().damping(18)</code>	0.15	62 %	1.8 s
<code>src/ui/TabBar.tsx:69</code> — active tab pill	<code>ZoomIn.springify().damping(16)</code>	0.13	66 % → peaks at scale 1.66	2.0 s
<code>src/ui/TabBar.tsx:82</code> — unread badge	<code>ZoomIn.springify().damping(14)</code>	0.12	69 % → peaks at scale 1.69	2.3 s
<code>src/ui/toast.tsx:76</code> — toast, travelling a full window height	<code>SlideInDown.springify().damping(18)</code>	0.15	62 %	1.8 s of its 3 s life
<code>src/app/puruka/[id].tsx:133</code> — photo pager dots	<code>LinearTransition.springify().damping(18)</code>	0.15	62 % → dot stretches to 28 px	1.8 s

$\zeta = \text{damping} / (2 \cdot \sqrt{\text{stiffness} \cdot \text{mass}})$; overshoot = $e^{(-\pi\zeta/\sqrt{1-\zeta^2})}$; settle $\approx 4/(\zeta\omega_0)$.

A second, quieter bug: `.duration(420).springify()` silently discards the duration. `springify(duration?)` assigns `this.durationV = duration` (`ComplexAnimationBuilder.ts:99–100`), so calling it with no argument overwrites the value with `undefined`. Every `.duration()` in the entering animations is dead code — the author did try to bound these, and the API threw it away. The working form is `springify(420)`.

Three further motion defects, unrelated to springs:

- **Skeleton shimmer** (`src/ui/index.tsx:393`) — `withRepeat(withTiming(1, {duration:1100, easing:Easing.inOut(Easing.ease)}), -1)` with no `reverse`, so the highlight decelerates to a near-stop at each edge and then snaps back. It reads as a pulse, not a sweep. The Puruka loading state runs **12 of these at once**. The highlight is `rgba(255,255,255,0.55)` — very hot in light mode.
- **ProgressBar** (`src/ui/index.tsx:439`) — 650 ms, and every loan's bar fires simultaneously on the Loans list.
- **Reduced motion** is handled only by accident. Reanimated's own default (`ReduceMotion.System`) covers its animations; the `Modal` fade in `PhotoViewer.tsx:105`, the `slide_from_right` stack transition

(`src/app/_layout.tsx:58`) and the `expo-image` cross-fades are not covered, and there is no in-app preference. Motion that is already correct and must be left alone: `FadeIn.duration(300)` on `EmptyState` and `ErrorView`; `SlideOutDown.duration(180)`; the `PhotoViewer` pinch/pan/zoom (180 ms); `expo-image transition={150|180}`; the native stack transition; platform spinners; `RefreshControl`.

2.2 Typography and Sinhala

Sinhala is the default language (`src/i18n/index.tsx:42`); English is opt-in.

`src/typography.tsx` is the best-informed file in the codebase. Its `useType()` hook returns a font family chosen by language and a line-height function, and its header comment explains exactly why: setting an `Inter` `fontFamily` on text containing Sinhala glyphs makes Android fall back to the **system** Sinhala face at regular weight, so bold silently disappears. Sinhala gets Noto Sans Sinhala at a 1.55 line-height ratio; Latin gets Inter at 1.35. Noto Sinhala has no 800 cut, so `extrabold` aliases to `Bold`.

That hook is not called in the places that need it most. There are 27 raw `fontWeight` usages across 14 files; 13 of those files never import `useType` at all:

<code>src/app/(auth)/login.tsx</code>	<code>src/app/(auth)/samithi.tsx</code>	<code>src/app/(tabs)/contributions.tsx</code>
<code>src/app/benefits.tsx</code>	<code>src/app/dues.tsx</code>	<code>src/app/guarantees.tsx</code>
<code>src/app/help.tsx</code>	<code>src/app/loan/[id].tsx</code>	<code>src/app/payouts.tsx</code>
<code>src/app/puruka-mine.tsx</code>	<code>src/app/puruka-new.tsx</code>	<code>src/app/requests.tsx</code>
<code>src/app/society.tsx</code>		

For a Sinhala-reading member — the default — every "bold" heading on those screens is not bold, and none of them sets a line height, so the glyph stack crowds between lines.

Sinhala also overflows in four known places:

Where	Constraint	The Sinhala string
Tab labels (<code>src/ui/TabBar.tsx:102–109</code>)	<code>fontSize:10.5</code> , <code>numberOfLines:1</code> , <code>maxWidth:76</code> , ~60 dp per tab on a 360 dp phone	මුල් පිටුව (10), දායක මුදල් (10), දැනුම්දීම (10) — all ellipsize
Home quick actions (<code>src/app/(tabs)/index.tsx:212–217</code>)	<code>fontSize:11</code> , <code>numberOfLines:2</code> , ~80 dp	සාමාජික කාඩ්පත
Theme segmented (<code>src/app/(tabs)/more.tsx:162–170</code>)	one non-scrolling row	පද්ධතිය / ආලෝක / අඳුරු
Money (<code>src/ui/index.tsx:201–208</code>)	fixed size, no <code>adjustsFontSizeToFit</code>	Rs. 1,250,000.00 beside a Sinhala label squeezes the label

The scale is not a scale. `src/theme.tsx:78` defines six sizes (`display 28` · `title 22` · `heading 17` · `body 15` · `caption 13` · `micro 12`) and is imported by exactly one file. Nineteen distinct sizes are actually in use, six of them fractional:

9.5 10.5 11 12 12.5 13 13.5 14 15 15.5 16 16.5 17 18 22 24 26 34 100

2.3 Colour and contrast

Computed with the WCAG 2.1 relative-luminance formula. AA requires 4.5:1 for body text.

Pair	Today	Result
primary <code>#3b82f6</code> as a link or icon on white	3.68:1	× fails
White label on a <code>#3b82f6</code> button	3.68:1	× fails

Pair	Today	Result
warning #d97706 on warningBg #fef3c7 (13 px)	2.86:1	× fails
success #16a34a on successBg #dcfce7 (13 px)	3.00:1	× fails
danger #dc2626 on dangerBg #fee2e2 (13 px)	3.95:1	× fails
textMuted #64748b on bg #f1f5f9	4.34:1	× fails
Body text #0f172a on surface #ffffff	17.85:1	✓

Three further colour defects:

- **The brand blue never adapts.** `primary`, `gradStart`, `gradEnd` and `navy` are byte-identical in the light and dark palettes (`src/theme.tsx:16–22` vs `:41–46`), so the gradient hero and every primary button are the brightest thing on a dark screen.
- **Three different blues ship.** `#1E64D4` in the app, `#1E6FD0` as both the Android adaptive-icon background and the notification accent (`app.json`). The notification shade shows a different blue from the app.
- `primarySoft #e8f0fe` **carries nine semantic roles** — tab pill, nav icon chips, row icon chips, quick actions, filter chips, category chips, secondary buttons, empty state, error state — so nothing reads as more important than anything else. Home alone shows five circular-chip sizes (30/34/44/46/72) in that one tint.
- The membership-card status badge hardcodes `#166534` / `#bbf7d0` / `#7f1d1d` / `#fecaca` (`src/ui/MembershipCard.tsx:74–75`) and so ignores dark mode entirely.

2.4 Structure and consistency

- **Zero** `StyleSheet.create` in the codebase; 370 inline style object literals.
- **Zero** `FlatList` or `SectionList`. Every list is `.map()` inside one `ScrollView` — including the infinite-scroll marketplace and a member's entire payment history.
- **Eight implementations of "a row inside a card"**, at six different vertical paddings (10/12/13/14/18 px): `more.tsx:71`, `more.tsx:112`, `contributions.tsx:62`, `payouts.tsx:35`, `benefits.tsx:29`, `loan/[id].tsx:80`, `society-funds.tsx:37`, `samithi.tsx:108`, `help.tsx:42`.
- **Four parallel status-badge colour maps:** `StatusBadge` (`src/ui/index.tsx:519`), `requests.tsx:17–26`, `puruka-mine.tsx:40–50`, `puruka/[id].tsx:58–66`.
- **Three chip styles**, two of them byte-identical copies (`notices.tsx:67–85`, `puruka.tsx:250–268`) and a third at a different radius and weight (`puruka-new.tsx:205–226`).
- **Three "big number in a box" variants** (`contributions.tsx:50`, `dues.tsx:38`, `society-funds.tsx:76`), the last bypassing `formatCurrency` and hand-building its `Rs. ...` string.
- **Six raw copies of the same error-text style**, while `Input` has an `error` prop that nothing uses.
- **Screen padding drifts 16 / 20 / 24** between the tab shell, New Request and the auth screens, so text left-edges jump across a push transition.
- **Token arithmetic defeats the grid:** `spacing.lg - 2`, `spacing.sm + 1`, `spacing.md - 2`, `radius.md - 4`.

2.5 Accessibility, as it stands

Concern	Today
<code>accessibilityLabel</code>	3 in the whole app; one of them is hardcoded English (<code>PhotoViewer.tsx:112</code>)
<code>accessibilityHint</code> , <code>accessibilityLiveRegion</code> , <code>hitSlop</code>	0 each
OS font scaling	No <code>allowFontScaling</code> , <code>maxFontSizeMultiplier</code> , <code>PixelRatio</code> or <code>fontScale</code> anywhere. Text does scale, but <code>lineHeight</code> is computed from a fixed design size and does not — at large sizes Sinhala ascenders clip first

Concern	Today
Touch targets	<code>Button</code> is 52 dp (good); <code>Segmented</code> options are ~34 dp (below the 44 dp minimum) — and that control is both the language and the theme switcher
Contrast	Six failing pairs, §2.3
Reduced motion	Accidental coverage only, §2.1

`MEMBER-MOBILE-APP-REQUIREMENTS.md:125` promises "large touch targets, dynamic font scaling, high contrast". Today that is an aspiration, not a fact. This document makes it true.

2.6 What is already good, and must survive

Worth stating plainly, because a refresh can destroy it by accident:

- `src/typography.tsx` — the language-aware type system, and the Android knowledge in its comments.
- `Card` 's deliberate removal of Android `elevation` (`src/ui/index.tsx:78`) because it smudges around large corner radii on real devices.
- The gradient-as-sibling pattern (`src/ui/index.tsx:236–245`) that keeps SVG from swallowing touches on Android — this cost two separate bug-fix commits.
- Always-labelled tabs, chosen because "many members are elderly and icon-only navigation loses them" (`src/ui/TabBar.tsx` header).
- `Money` always rendering in Inter with `tabular-nums` , so amounts align.
- A single, honest token module with a comment admitting its own partial adoption.
- Light/dark parity across every shared component.
- No emoji used as iconography anywhere.

3. The design system

3.1 Principles

1. **Motion explains, never entertains.** Every animation answers "what just happened?" or "where did that come from?". Nothing bounces. (Taken verbatim from the web app, and the direct answer to the owner's complaint.)
2. **Legible before elegant.** When density and readability conflict, readability wins. The reader may be 70 and in sunlight.
3. **Sinhala is the design language, not a translation.** Type, line height, label length and layout are designed for Sinhala first and checked in English second.
4. **One eSamithi.** A member's app and the officer's desktop are visibly the same product.
5. **The token is the only number.** No literal spacing, size, radius or colour in a screen file. No arithmetic on tokens.
6. **Calm surfaces.** One accent colour, one card shape, one row height, generous whitespace, no decoration that does not carry meaning.

3.2 Motion — the core of this refresh

A new module, `mobile/src/motion.ts`, is the only place motion values exist. **Rule: no call site may pass a partial spring config.** Every spring states `damping`, `stiffness` and `mass`, because omitting `mass` is the entire bug described in §2.1.

```
// mobile/src/motion.ts
import { Easing, ReduceMotion } from 'react-native-reanimated'

// Reanimated 4's own default is critically damped (damping 120, mass 4,
// stiffness 900 -> zeta = 1.00, ~267ms). These tokens keep that character and
// state every field, so no future edit can half-override one and reintroduce
// the wobble this module was written to remove.
export const SPRING = {
  /** Screen and section entrances. zeta = 1.00, settles ~210ms. */
  enter: { damping: 38, stiffness: 360, mass: 1, reduceMotion: ReduceMotion.System },
  /** Anything that should feel physical but must never overshoot. zeta = 1.00, ~267ms. */
  settle: { damping: 120, stiffness: 900, mass: 4, reduceMotion: ReduceMotion.System },
} as const

export const TIMING = {
  press: { duration: 90, easing: Easing.out(Easing.quad) },
  fast: { duration: 150, easing: Easing.out(Easing.quad) },
  base: { duration: 220, easing: Easing.out(Easing.cubic) },
  page: { duration: 320, easing: Easing.out(Easing.cubic) },
} as const

/** First-mount stagger only: 30ms per item, at most 3 items. */
export const stagger = (i: number): number => Math.min(i, 2) * 30
```

Token table — as built. Every value computed, not estimated.

Token	Value	ζ	Settle	Used for
<code>dur.pressIn / dur.pressOut</code>	<code>withTiming 90 / 140 ms, ease.press in, ease.standard out</code>	—	90 / 140 ms	Every tap. Asymmetric on purpose: the finger drives the press down, the release settles. A curve, not a spring — structurally incapable of wobbling
<code>dur.micro</code>	150 ms	—	150 ms	Tab pill, unread badge, chips, pager dots
<code>dur.exit</code>	160 ms	—	160 ms	Every exit — deliberately shorter than its entrance
<code>dur.content</code>	160 ms	—	160 ms	Skeleton giving way to content, on <code>Screen</code>
<code>dur.gesture</code>	180 ms	—	180 ms	Photo zoom/pan and image cross-fades. Already correct; tokenised, not changed
<code>dur.surface</code>	220 ms	—	220 ms	Toast, banner
<code>dur.enter</code>	260 ms	—	260 ms	Welcome-screen arrivals, empty and error states
<code>dur.page</code>	320 ms	—	320 ms	Progress fill. Also roughly the native stack transition, which is left alone
<code>dur.stagger</code>	30 ms per item	—	—	Welcome screen only
<code>dist.rise</code>	12 px	—	—	How far an entering element travels. Short, because <code>ease.enter</code> is front-loaded
<code>SPRING.settle</code>	<code>{damping:120, stiffness:900, mass:4}</code>	1.00	~390 ms	The only spring left in the app. Reanimated 4's own Gentle numbers, stated in full

Easings: `ease.standard bezier(.2,0,0,1) · ease.enter bezier(0,0,0,1) · ease.exit bezier(.3,0,1,1) · ease.press Easing.out(Easing.quad) · ease.pulse Easing.inOut(Easing.quad)`. This is the web app's 80/150/220/320 ladder (`ESAMITHI-WEB-APP-REQUIREMENTS.md` §4.5); mobile's press token is 90 ms rather than 80 ms because a touch has no hover state to precede it.

Bezier easings are `.factory()` **'d once at module scope**. Built per render they allocate a new worklet each time, and an un-factored curve handed to a layout builder trips `assertEasingIsWorklet` in dev and misbehaves silently in release.

Per-interaction, as built.

Interaction	Was	Now	Where
Any press	spring ζ 0.25, 44 % overshoot, 1.6 s	90 ms down, 140 ms back	<code>src/ui/pressable.tsx</code>
Tab switch	pill <code>ZoomIn</code> peaking at 1.66, 2.0 s	<code>FadeIn</code> 150 ms. Not a duration-solved spring: that solver runs a 100-iteration bisection on the UI thread at every start, and this fires on every press	<code>src/ui/TabBar.tsx</code>
Unread badge	<code>ZoomIn</code> peaking at 1.69, 2.3 s	<code>FadeIn</code> 150 ms. No pop, ever — this badge counts death notices	<code>src/ui/TabBar.tsx</code>
Home opens	7-section cascade, still moving after 2.2 s	The cascade is deleted . Home is a tab screen that expo-router keeps mounted, so it fired once per launch and never again	<code>src/app/(tabs)/index.tsx</code>
Skeleton → content, every screen	a hard cut — the skeleton <code>Screen</code> unmounts and a content <code>Screen</code> mounts	one 160 ms crossfade, moved onto <code>Screen</code> so all 18 screens get it	<code>src/ui/index.tsx</code>
Welcome	5 elements, 62 % overshoot each, ~2.1 s	5 elements, 260 ms each, 30 ms apart, 12 px rise — 380 ms total. The one stagger that earns its keep	<code>src/app/(auth)/index.tsx</code>

Interaction	Was	Now	Where
Toast	<code>SlideInDown</code> sprang the layout <code>originY</code> a full window height; flat 180 ms exit	<code>FadeInDown</code> 220 ms in, <code>FadeOutDown</code> 160 ms out — symmetric, and it rises into place because it is anchored at the bottom	<code>src/ui/toast.tsx</code>
Pager dots	<code>LinearTransition</code> spring — one swipe shuffled the whole row for 1.8 s	fixed 20 × 7 slot, inner bar <code>scaleX</code> 0.35 ↔ 1.0 at 150 ms. Nothing changes layout, so nothing else moves	<code>src/app/puruka/[id].tsx</code>
Skeleton	12 independent 1100 ms sweeps, <code>Easing.inOut</code> and no <code>reverse</code> , so a white bar snapped back each cycle	one shared 900 ms opacity breath, <code>reverse: true</code> , in phase everywhere	<code>src/ui/index.tsx</code>
Progress fill	650 ms, and <code>scaleX</code> squashed the fill's own rounded cap	320 ms; the radius moved to the track	<code>src/ui/index.tsx</code>
Stack push · photo zoom · empty/error fade	already correct	unchanged — tokenised and reduced- motion gated only	—

Banned outright, in this codebase, permanently:

- `.springify()` with no argument after a `.duration()` — it discards the duration. Use `springify(220)` .
- Any `withSpring` config that omits `mass` .
- `ZoomIn` / `ZoomOut` on anything that is not a photo viewer.
- `withRepeat` on anything except the skeleton shimmer.
- Any animation over 400 ms that is not a stack transition.
- Stagger beyond 3 items, or on anything but a first mount.

Reduced motion is explicit. `reduceMotion: ReduceMotion.System` is stated on every token rather than inherited silently, and the three things Reanimated cannot see — the `PhotoViewer Modal` fade, the native stack transition and the `expo-image` cross-fades — are gated on `useReducedMotion()` , re-exported from the same module.

3.3 Typography

One scale, six sizes, no fractionals. The body rises 15 → 16 px (decision D3).

Token	Size	Weight	Line height (Latin ×1.35)	Line height (Sinhala ×1.55)	Used for
<code>display</code>	28	Bold	38	43	The one big number on a screen (total owed, total contributed)
<code>title</code>	22	Bold	30	34	Screen titles, the Home greeting
<code>heading</code>	18	SemiBold	24	28	Card titles, section headings
<code>body</code>	16	Regular / SemiBold	22	25	Everything a member reads
<code>label</code>	14	SemiBold	19	22	Row labels, captions, metadata
<code>micro</code>	12	SemiBold	16	19	Badges, tab labels, timestamps

Rules:

1. **Every Text gets its family from** `useType()` . No exceptions, no `fontWeight` outside `src/typography.tsx` . This single change fixes bold for the majority of users.
2. **Every Text gets a line height** from `ty.lh(size)` .
3. **Money is always Inter with** `tabular-nums` , never the Sinhala face, so columns align (already true in `Money` — keep it).

4. **Line height follows the scaled size**, not the design size: `ty.lh(size * PixelRatio.getFontScale())`, so Sinhala ascenders do not clip when a member raises their phone's text size.
5. `maxFontSizeMultiplier` is set only where layout genuinely cannot reflow — tab labels (1.3) and badges (1.2). Nowhere else.
6. Sinhala never exceeds weight 700 (Noto has no 800 cut).

3.4 Colour

Adopting the web app's scale (decision D2) fixes six AA failures at the same time. All ratios below are computed.

Light

Token	Today	Proposed	Check
primary	#3b82f6	#1E64D4	text/icon on white 3.68 → 5.48:1 ✓
onPrimary on primary	white, 3.68:1 ×	white on #1E64D4	5.48:1 ✓
primarySoft	#e8f0fe	#EEF4FE	primary on it 4.96:1 ✓
textMuted	#64748b, 4.34:1 ×	#475569	on bg 6.92:1 ✓
success	#16a34a, 3.00:1 ×	#166534	on successBg 6.49:1 ✓
warning	#d97706, 2.86:1 ×	#B45309	on warningBg 4.51:1 ✓
danger	#dc2626, 3.95:1 ×	#B91C1C	on dangerBg 5.30:1 ✓
bg / surface / surfaceAlt / text	#f1f5f9 / #ffffff / #f8fafc / #0f172a	unchanged	body text 17.85:1 ✓

Dark — the brand must finally adapt (§2.3):

Token	Today	Proposed	Check
primary	#3b82f6 (same as light)	#4C8DF6	on surface 5.11:1 ✓
onPrimary	#ffffff, 3.68:1 ×	#0F172A (navy on the lighter blue)	5.48:1 ✓
primarySoft	#1e2f55	#1B2B4D	chip text uses #85B0F2 → 6.33:1 ✓
textMuted	#94a3b8	unchanged	on bg 7.30:1 ✓
success / warning / danger	#4ade80 / #fbbf24 / #f87171	unchanged	8.55 / 8.97 / 5.84:1 ✓

Gradient. `gradStart #3b82f6 → gradEnd #1E64D4` in light. In dark it becomes `#2E75E3 → #1854B8`, so the hero stops being the brightest object on the screen.

`primarySoft` **gets one job: the selected state of a control** (tab pill, selected chip, secondary button). Everything that used it as decoration — nav icon chips, row icon chips, quick actions, empty state, error state — moves to `surfaceAlt`. This is what restores hierarchy: when only selection is blue, selection is visible.

Icon chips get two sizes, not five: 40 dp inside a row, 56 dp for an empty or error state. The 30/34/44/46/72 spread on Home disappears.

Membership card badge stops hardcoding four hex values (`src/ui/MembershipCard.tsx:74–75`) and uses `success / successBg` and `danger / dangerBg` so it follows the theme.

3.5 Spacing, radius, elevation

Spacing keeps the 4-pt grid and **gains the steps people were faking** with arithmetic, so `spacing.lg - 2` has a name:

```
spacing = { xs:4, sm:8, md:12, lg:16, xl:20, xxl:24, xxxl:32 } // unchanged
radius  = { sm:8, md:12, lg:16, xl:24, pill:999 }           // aligned to the web app
```

- Arithmetic on a token is banned. If a value is needed that the scale lacks, the scale is wrong — add the step.
- **One screen padding:** `spacing.lg (16)`. The 16/20/24 drift between the tab shell, New Request and the auth screens ends; text left-edges no longer jump across a push.
- **One card:** `radius.lg`, `spacing.lg` padding, 1 px hairline, iOS-only shadow (`elevation.sm`), Android `elevation` stays off — see §2.6.
- **One row height:** 56 dp minimum, replacing the six current values.
- Elevation is unchanged and already well-behaved.

3.6 Components

New or reworked in `mobile/src/ui/`. Each one exists to delete duplication found in §2.4.

Component	Replaces	Notes
<code>ListRow</code>	8 hand-rolled row implementations at 6 heights	Icon chip (40 dp, optional) · label · value · chevron (optional) · separator handling · <code>onPress</code> . Minimum 56 dp. The single most duplicated pattern in the app
<code>StatusPill</code>	4 parallel colour maps (<code>StatusBadge</code> , <code>requests.tsx:17</code> , <code>puruka-mine.tsx:40</code> , <code>puruka/[id].tsx:58</code>)	One status vocabulary, one colour map, every value translated, a leading dot so state survives colour blindness
<code>Chip</code>	3 chip styles, two of them copy-paste	Selected / unselected, one radius, one height (36 dp), <code>accessibilityState.selected</code>
<code>AmountCard</code>	3 "big number in a box" variants	Label + <code>Money</code> at <code>display</code> , optional tone; always via <code>formatCurrency</code>
<code>Input</code> (rework)	6 raw copies of the same error text	Adopt the existing unused <code>error</code> prop. Filled background (<code>surfaceAlt</code>) so the border is not load-bearing; focused border <code>primary</code> (5.48:1 light / 5.11:1 dark)
<code>Card</code> (rework)	screens overriding <code>paddingVertical</code> with raw numbers	Explicit <code>padded</code> / <code>list</code> variants
<code>Screen</code> (rework)	—	Gains a <code>FlatList</code> mode so long lists stop rendering into one <code>ScrollView</code>
<code>Segmented</code> (rework)	—	Minimum 48 dp tall (from ~34), and it wraps or scrolls rather than crushing Sinhala

3.7 Lists

`FlatList` replaces `.map()` wherever the list is unbounded: Contributions (a member's whole payment history), Notices (the whole archive), Puruka feed (paged), Payouts, Requests, Guarantees. `keyExtractor`, `getItemLayout` where rows are fixed, and `ListEmptyComponent` wired to the existing `EmptyState`.

4. Screen specifications

Only the screens whose composition changes are specified here. Every other screen inherits the new tokens, type, motion and components without a layout change — that is the point of decision D1.

4.1 Home (`src/app/(tabs)/index.tsx`) — the priority

Today it is seven unrelated card shapes in one scroll: a greeting with a 44 dp gradient avatar, a compact membership card, a conditional dues banner, a notice card with a coloured left border, two stat cards with 34 dp chips, a four-item quick-action row with 46 dp chips and no container, and an activity list card. Section headings appear above only three of them, and the dues heading renders conditionally (:130), so the page's rhythm changes depending on whether the member owes money. Eight distinct font sizes appear on this one screen.

Required composition — one question answered per block, in the order a member asks them:

1. **Header.** Greeting at `title`, member ID and join date at `label / muted`, avatar 40 dp. Not a section — it is the page's own title.
2. **Status.** The answer to "do I owe anything", always present, never conditional: one card, `danger / warning / success` tone, the amount at `display` when something is owed, tappable through to Dues. This replaces the conditional banner stack and fixes the rhythm problem.
3. **Membership card.** Unchanged in content; it is the emotional anchor and members show it at the office.
4. **Two figures.** Total contributed · loan balance, as `AmountCard` s. Icon chips 40 dp, `surfaceAlt` not `primarySoft`.
5. **Latest notice**, if any. One card, type colour on the icon and label only — the 4 px left border goes; it is the only element in the app using that device.
6. **Quick actions.** Four `ListRow` s in one card rather than four unlabelled circles — at 16 px Sinhala, a label under a 46 dp circle cannot fit (§2.2) and a row can.
7. **Recent activity.** Five `ListRow` s in one card.

Every block gets a `SectionHeader`, or none does. Stagger is limited to the first three blocks (§3.2).

4.2 Dues (`src/app/dues.tsx`)

The most important screen in the app after Home, and today it is a plain stack of cards. Required: total owed as a single `AmountCard` at `display`; then one card per overdue loan using `ListRow` for the principal/interest/fine breakdown; then the membership-fee section; then the existing empty state when the member is clear. No layout invention — just the new row, the new amount treatment and the new type scale.

4.3 Contributions (`src/app/(tabs)/contributions.tsx`)

Convert to `FlatList` with month headers (`SectionList` is acceptable here). Summary `AmountCard` becomes the list header. Rows become `ListRow`. This screen currently renders a member's entire payment history into one `ScrollView` — after several years of monthly dues that is the app's first real performance ceiling.

4.4 Loans (`src/app/(tabs)/loans.tsx`) and loan detail (`src/app/loan/[id].tsx`)

Status as `StatusPill`. Amounts through `AmountCard / Money` — the detail screen currently hand-builds repayment subtitles with `/100 .toLocaleString()` (`loan/[id].tsx:95–96`), which drops the `Rs.` prefix and the two-decimal rule every other screen follows; route it through `formatCurrency`. Progress bar ≤ 400 ms, and on the list the bars animate only on first mount, not on every re-render.

4.5 Notices (`src/app/(tabs)/notices.tsx`)

Filter chips become `Chip`. Notice cards keep the type icon and label but drop the 4 px coloured left border in favour of a tinted icon chip, so they match every other card in the app. `FlatList`.

4.6 Tab bar (`src/ui/TabBar.tsx`)

The active indicator **slides** rather than zooming (§3.2). Labels stay — they are a deliberate accessibility decision (§2.6) — but the three long Sinhala labels must be shortened to fit ~60 dp at `micro`, and **this needs a Sinhala speaker's sign-off, not a developer's guess** (§9, open decision O1). `maxFontSizeMultiplier: 1.3`.

4.7 Welcome (`src/app/(auth)/index.tsx`)

`:23–27` redirects to the samithi-code screen whenever there is no pending profile, so a first-time member never sees the Welcome screen — the brand moment, the logo, the tagline — until they come back from entering a code. Reverse the order: Welcome first, then the code screen. This is a two-line change and it is the app's first impression.

4.8 Puruka new/edit (`src/app/puruka-new.tsx`)

Five validation failures currently raise `Alert.alert('', message)` — an OS dialog with an empty title (`:101–117`) — while every other form in the app shows inline text. Use the `Input` `error` prop.

5. Accessibility requirements

WCAG 2.1 AA is the bar, adapted to a native app and to this audience.

#	Requirement	How it is checked
A1	Every text pair reaches 4.5:1; every control boundary and state indicator 3:1	Script over the palette (§8)
A2	Every interactive element ≥ 48 dp in both dimensions	Review; <code>Segmented</code> and the Puruka clear-search button are the known failures
A3	Every icon-only control has a translated <code>accessibilityLabel</code>	<code>grep</code> — today there are 3 labels app-wide, one hardcoded English
A4	<code>ListRow</code> announces label and value as one unit (<code>accessible</code> , composed label)	Screen-reader pass on Home and Dues
A5	Text scales to 200 % without clipping or loss of function; line height scales with it	Device pass at maximum OS font size, in Sinhala
A6	Reduced motion honoured explicitly, including <code>Modal</code> , stack and image transitions	Device pass with the OS toggle on
A7	Colour is never the only carrier of meaning	<code>StatusPill</code> keeps its leading dot and gains a distinct label per state
A8	Toasts and scan feedback announce via a live region	Screen-reader pass
A9	Every new string exists in both <code>en.ts</code> and <code>si.ts</code>	Parity check in CI (§8)

6. Non-functional requirements

#	Requirement
N1	Cold start no slower than today on a low-end Android; the seven font files (1.03 MB) stay as they are
N2	No unbounded list renders into a <code>ScrollView</code> (§3.7)
N3	Frame rate stays at 60 fps while scrolling Contributions with 5 years of history on a budget device
N4	The refresh adds no new npm dependency — everything specified here uses what is already installed
N5	Phases 1–5 ship over the air; nothing in them touches <code>app.json</code> <code>plugins</code> , the icon or the splash
N6	<code>npx tsc --noEmit</code> green at every commit; typed routes unchanged
N7	React Compiler stays enabled; new components obey the rules of React

7. Delivery

Six phases. **Each one is independently shippable** — this matters because there is no test suite to catch a half-finished refresh (§10).

Phase	Scope	Ships	Exit criteria
0 • This document	Approval of the direction	—	Owner signs off §1.2
1 • Motion  done	<code>src/motion.ts</code> ; every call site in <code>pressable.tsx</code> , <code>TabBar.tsx</code> , <code>toast.tsx</code> , <code>(tabs)/index.tsx</code> , <code>(auth)/index.tsx</code> , <code>puruka/[id].tsx</code> , <code>Screen</code> , <code>Skeleton</code> , <code>ProgressBar</code> , <code>PhotoViewer</code> , <code>_layout.tsx</code> ; explicit reduced motion; <code>scripts/check-motion.mjs</code> wired into CI	OTA	 No spring remains outside <code>SPRING.settle</code> ($\zeta = 1.00$); the gate fails on the original bug when reintroduced; <code>tsc</code> green; bundle exports and runs with no runtime errors. Outstanding: the owner confirms on a preview build that nothing dances
2 • Sinhala and type	<code>useType()</code> everywhere; delete 27 <code>fontWeight</code> s; six-size scale; tab labels; font scaling	OTA	No <code>fontWeight</code> outside <code>typography.tsx</code> ; Sinhala headings bold on a real Android; 200 % font scale clean
3 • Colour	Web brand scale, AA-passing semantics, dark-mode primary and gradient, membership-card badge	OTA	Contrast script passes in both themes
4 • Components	<code>ListRow</code> , <code>StatusPill</code> , <code>Chip</code> , <code>AmountCard</code> ; <code>Input</code> / <code>Card</code> / <code>Screen</code> / <code>Segmented</code> rework; retire duplicates	OTA	Duplicate implementations deleted, not merely unused
5 • Screens	Home, Dues, Contributions, Loans, Notices; <code>FlatList</code> ; Welcome reachable	OTA	Before/after screenshots in four states (§8)
6 • Binary (deferred)	Unify the three blues; icon, splash, adaptive icon, notification colour; new Play Store screenshots; versionCode 7	Store	Owner schedules it

Sequencing rationale: motion first because it is the loudest complaint and the smallest diff; Sinhala second because it affects the majority of users; colour third because it is self-contained; components before screens because screens consume them.

8. Verification

CI runs only `npx tsc --noEmit`. There are no tests, no lint config and no visual regression, so verification has to be deliberate and mostly scripted.

Automated, added by this work (`mobile/scripts/` , wired into `clients.yml`). `check-motion.mjs` exists and runs in CI as of phase 1; the other three arrive with their phases:

Check	Fails when
<code>check-motion.mjs</code>	Any <code>withSpring</code> config omits <code>mass</code> , any <code>.springify()</code> has no argument, any computed $\zeta < 0.9$, any duration > 400 ms outside the allow-list
<code>check-contrast.mjs</code>	Any palette text pair < 4.5:1, any control boundary or state indicator < 3:1, in either theme
<code>check-type.mjs</code>	Any <code>fontWeight</code> outside <code>src/typography.tsx</code> , any <code>fontSize</code> not on the six-step scale, any <code>Text</code> without a line height
<code>check-i18n.mjs</code>	<code>en.ts</code> / <code>si.ts</code> key parity broken, or a <code>t('...')</code> key missing from both

These are the same arithmetic used to produce §2.1 and §2.3, so the document and the gate cannot drift apart.

By hand, on a real Android device — the matrix is four states, and Sinhala is checked first:

	Light	Dark
Sinhala (default)	✓ every screen	✓ every screen
English	✓ every screen	✓ every screen

Plus: OS font size at maximum; OS reduce-motion on; a 360 dp phone (the narrowest common Sri Lankan Android); and a screenshot set of Home, Dues, Contributions, Loans and Notices before and after, attached to this document as an appendix.

By the owner: a `preview` -channel EAS build, opened and used. The acceptance question is his own — *does it still dance?*

9. Open decisions

#	Decision	Default if not answered
01	The three long Sinhala tab labels (මුද් ජිවුට් , දායක මුදල් , දැනුම්දීම) must shorten to fit ~60 dp. A Sinhala speaker must choose the words — this is not a developer's call	Phase 2 ships with the labels unchanged and allowed two lines, which is worse-looking but not wrong
02	Should the Home "Status" block show the exact amount owed, or only a state? Showing money on the first screen is a privacy consideration in a shared household	Show the amount; it is the question the app exists to answer
03	Phase 6 timing — the icon, splash and notification blue are visibly a different blue from the app until it ships	Bundle it with the next feature release rather than a release of its own
04	Whether to keep the gradient at all. It is used in 7 places and is the app's only decorative device; a flatter treatment would be more "banking app" and less distinctive	Keep it, with the dark-mode variant from §3.4

10. Risks

Risk	Mitigation
No safety net. No tests, no visual regression. A sweep across 14 files can break a screen silently	Phase the work; ship each phase separately; before/after screenshots in four states; the four scripted gates in §8
Android has bitten this codebase three times — SVG gradients swallowing touches, <code>elevation</code> smudging large radii, Sinhala bold falling back to the system face	§2.6 records all three; the gates in §8 catch the third mechanically; the first two are code-review rules
Raising the type floor fits less per screen	Intended (D3), but check on a 360 dp phone before phase 5 lands
<code>ListRow</code> is used on every screen — a bug in it is a bug everywhere	Land it in phase 4 on its own, ship, then migrate screens in phase 5
OTA means a bad release reaches every member within a day	Each phase goes to the <code>preview</code> channel first and is opened by the owner before the <code>prod</code> channel
Sinhala judgement cannot be made by the implementer	O1 is explicitly the owner's

Appendix A · Files this work touches

New: `src/motion.ts` · `src/ui/ListRow.tsx` · `src/ui/StatusPill.tsx` · `src/ui/Chip.tsx` · `src/ui/AmountCard.tsx` · `scripts/check-motion.mjs` · `scripts/check-contrast.mjs` · `scripts/check-type.mjs` · `scripts/check-i18n.mjs`

Reworked: `src/theme.tsx` (palette, radius) · `src/typography.tsx` (scaled line height) · `src/ui/index.tsx` (Card, Input, Screen, Segmented, Skeleton, ProgressBar, Money, SectionHeader) · `src/ui/pressable.tsx` · `src/ui/TabBar.tsx` · `src/ui/toast.tsx` · `src/ui/MembershipCard.tsx`

Screens re-composed: `src/app/(tabs)/index.tsx` · `src/app/dues.tsx` · `src/app/(tabs)/contributions.tsx` · `src/app/(tabs)/loans.tsx` · `src/app/loan/[id].tsx` · `src/app/(tabs)/notices.tsx` · `src/app/(auth)/index.tsx` · `src/app/puruka-new.tsx`

Token/type sweep only (no layout change): the remaining 20 screen files, principally the 13 listed in §2.2.

Deferred to phase 6: `app.json` (adaptive icon `#1E6FD0`, notification colour, splash) · `assets/images/*` · `store-assets/*`

Appendix B · Reference

Thing	Where
What the app does	<code>docs/MEMBER-MOBILE-APP-REQUIREMENTS.md</code>
The shared design system	<code>docs/ESAMITHI-WEB-APP-REQUIREMENTS.md</code> §4
Expo version pin	<code>https://docs.expo.dev/versions/v57.0.0/</code> — <code>mobile/AGENTS.md</code> requires reading the versioned docs before writing code
Reanimated spring defaults	<code>node_modules/react-native-reanimated/src/animation/spring/springConfigs.ts</code>
Build profiles	<code>mobile/eas.json</code> — <code>preview</code> (testbed channel) and <code>production</code> (prod channel)
Regenerate this document's PDF	<code>node docs/build-pdf.mjs ESAMITHI-MOBILE-UIUX-REQUIREMENTS</code>